# Mathematical Methods for International Commerce

## Week 2/1: Graphs of Equations, Solving Equations

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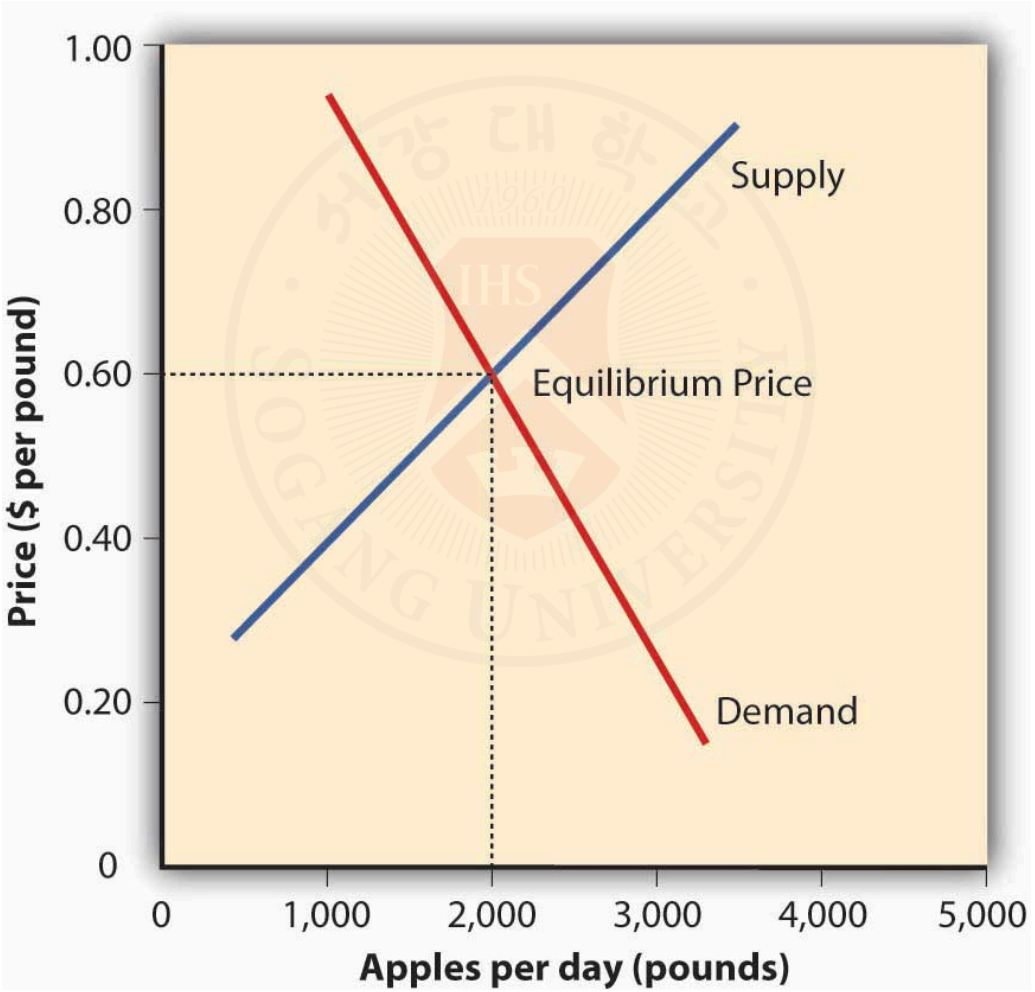
# Why Are These Concepts Essential in Economics?

## Supply and Demand Graph



# Why Are These Concepts Essential in Economics? (cont)

## Visualizing Economic Relationships

- **Supply and Demand Curves**: Graphs illustrate how prices and quantities interact in markets.
- **Cost and Revenue Functions**: Visual tools to analyze profitability and break-even points.

## Solving Equations for Economic Analysis

- **Equilibrium Analysis**: Determining market-clearing prices and quantities.
- **Optimization**: Maximizing profit or minimizing cost functions.

## Real-World Applications

- **Policy Modeling**: Predicting outcomes of fiscal and monetary policies.
- **Business Strategy**: Informing pricing, production, and investment decisions.

*Let's dive in!*

# Learning Objectives

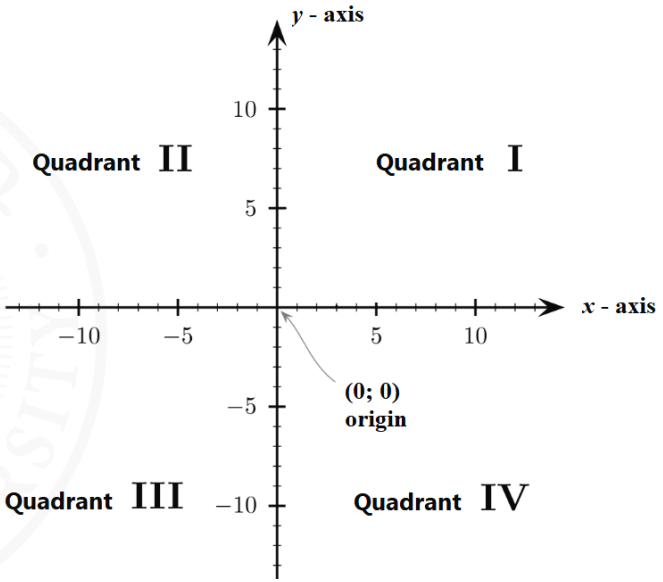
At the end of this class, you should be able to:

- **Graph** linear equations in one/two variables.
- **Solve** a linear equation and a system of two simultaneous linear equations using **elimination**.
- **Detect** when a system of equations **has no solution**.
- **Identify** when a system of equations has **infinitely many solutions**.
- **Solve** a system of **three equations with three unknowns** using **elimination**.

# Section 1.3: Graphs of Equations

## Understanding the Cartesian Coordinate System

- **X-axis**: Horizontal line
- **Y-axis**: Vertical line
- **Origin**: Intersection of X and Y axes (0,0)
- **Quadrants**: Four sections divided by the axes



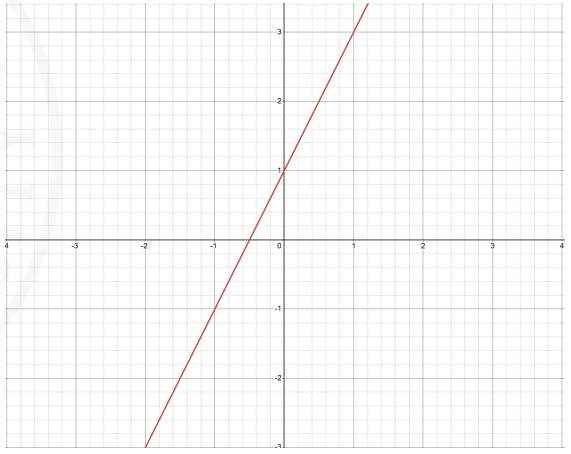
# Plotting Linear Equations

## Understanding Linear Equations

- **Linear Equation**: $y = mx + c$
- $m$: **Slope of the line**
- $c$: **Y-intercept** (point where the line crosses the Y-axis)

**Example:**

- Plot the equation ( y = 2x + 1 ).



Source: Desmos Graphing Calculator

# Your turn

## Please plot the following linear equations:

1. $(y = 3x - 2)$
2. $(y = -2x + 3)$
3. $(y = 0.5x + 1)$

# Impact of Exchange Rate on Loan Payments

## How Exchange Rate Affects Loan Repayments

- A company operates in Eurozone, so it cashes out in **EUR**.
- A company must pay **$200,000 annually** for a loan (cash outflow).
- The payment in **EUR depends on the exchange rate**.
- If the **exchange rate changes**, the **amount in EUR** fluctuates.
  - Say, exchange rate fluctuates from **€1 = $0.90** to **€1 = $1.20**.
- There is a fixed commission fee of **€50,000** for the transaction.

# Debt and Exchange Rate Relationship

## Equation for Debt Calculation

A company borrows in **USD**, but its total debt in **EUR** depends on the exchange rate and a **fixed commission**.

$$D = \frac{L}{ER} + C$$

Where:

- D = Total debt in **EUR**
- L = Loan amount in **USD**
- ER = Exchange rate **(USD per EUR, USD/EUR)**
- C = Fixed commission **(€50,000 EUR)**

## Example Calculation

A company borrows **$1,000,000 USD** (principal/obligation), and the exchange rate is **1 EUR = 1.10 USD**, with a **€50,000 commission**.

$$D = \frac{1,000,000}{1.10} + 50,000$$

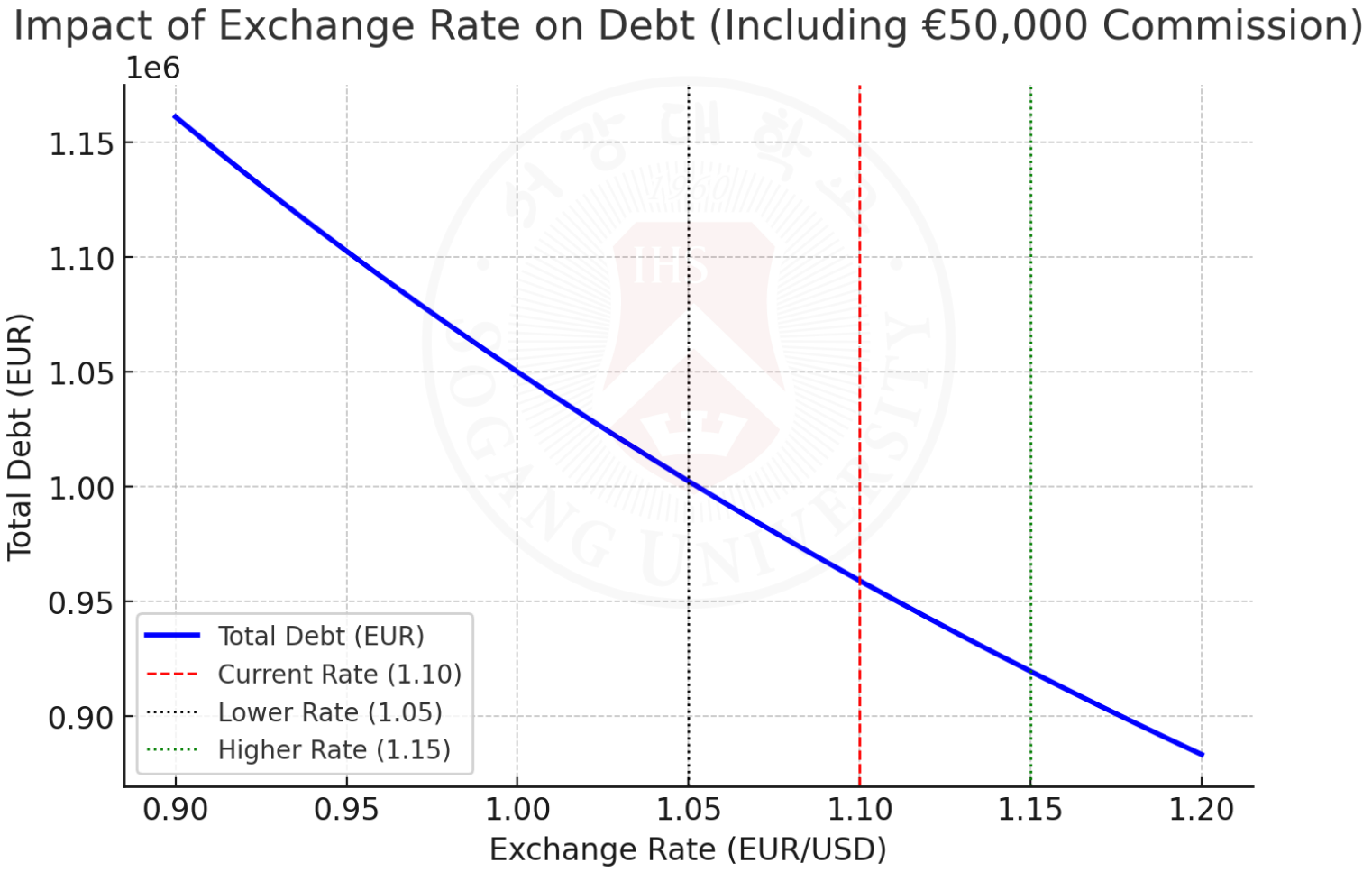
$$D = 909,091 + 50,000 = 959,091 \text{ EUR}$$

If the **exchange rate drops** to **1.05 EUR/USD**, the debt increases:

$$D = \frac{1,000,000}{1.05} + 50,000 = 952,381 + 50,000 = 1,002,381 \text{ EUR}$$

**As the EUR weakens, the debt in EUR increases even more!**

# Visualizing Impact of Exchange Rate on Loan Payments (cont)



# What is a Simultaneous Equation?

- A **system of equations** consists of two or more equations with multiple variables.
- The goal is to **find the values of the unknowns** that satisfy all equations simultaneously.

Example of a system with two unknowns:

$$2x + 3y = 12$$
$$4x - y = 5$$

# Method: Solving by Elimination (Two Equations)

**Steps to solve**:

1. **Multiply or adjust** the equations to align one variable.
2. **Add or subtract** the equations to eliminate one variable.
3. **Solve** for the remaining variable.
4. **Substitute** back to find the second variable.

**Example: Solve the system**

$$3x + 2y = 14$$
$$5x - 2y = 10$$

**Step 1: Add the two equations**

$$(3x + 2y) + (5x - 2y) = 14 + 10$$

$$8x = 24 \Rightarrow x = 3$$

## Example: Solve the system (cont)

### Step 2: Substitute ( x = 3 ) into one equation

$$3(3) + 2y = 14$$

$$9 + 2y = 14 \Rightarrow 2y = 5 \Rightarrow y = 2.5$$

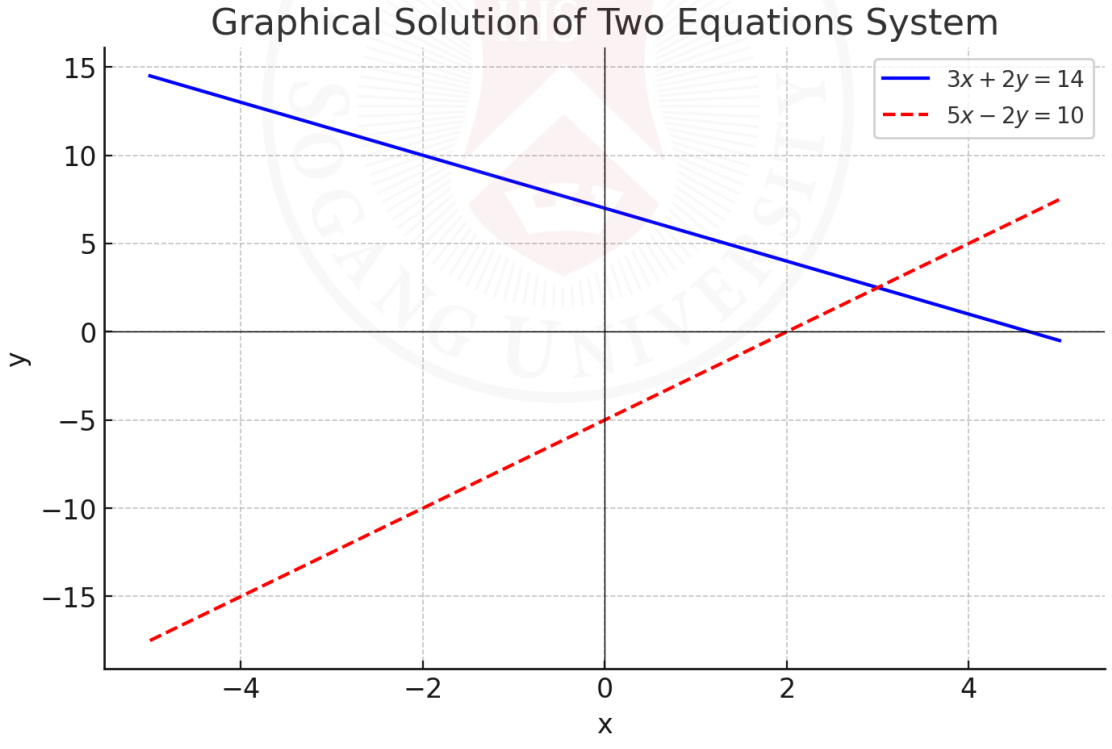
**Solution**: ( x = 3, y = 2.5 )

# Graphical Interpretation of a System of Equations

## Graphical Solution

- **Graph each equation** on the same axes.
- **Intersection point** is the solution to the system.

**Example:**



# No Solution Case (Inconsistent System)

A system has **no solution** if the equations are contradictory.

**Example:**

$$2x + 4y = 10$$
$$x + 2y = 6$$

Dividing the first equation by 2:

$$x + 2y = 5$$

This contradicts the second equation **( x + 2y = 6 )**!

**No solution exists** → The lines are **parallel**.

# Infinitely Many Solutions (Dependent System)

A system has **infinitely many solutions** if the equations are identical.

**Example:**

$$2x + 3y = 6$$
$$4x + 6y = 12$$

Divide the second equation by **2**:

$$2x + 3y = 6$$

**The equations are identical**, meaning **infinitely many solutions** exist.

**Graphically, the lines overlap completely**.

# Your Turn: Practice Problems

## Please solve the following systems of equations:

1. ( $x + 2y = 5$ ) and ( $2x - y = 3$ )
2. ( $3x + 2y = 14$ ) and ( $5x - 2y = 10$ )
3. ( $2x + 4y = 10$ ) and ( $x + 2y = 6$ )

# Solving a 3×3 System Using Elimination

## Problem: Solve the System

$$x + y + z = 6 \quad (1)$$
$$2x - y + 3z = 14 \quad (2)$$
$$x + 2y - z = 4 \quad (3)$$

# Step 1: Eliminate ( z )

Adding **(1) and (3)**:

$$2x + 3y = 10 \quad (4)$$

Multiplying **(1) by -2** and adding to (2):

$$-3y + z = 2 \quad (5)$$

# Step 2: Express ( z ) in Terms of ( y )

From (5):

$$z = 3y + 2$$

Substituting into (1):

$$x + y + (3y + 2) = 6$$

$$x + 4y = 4 \quad (6)$$

Now solve:

$$2x + 3y = 10 \quad (4)$$
$$x + 4y = 4 \quad (6)$$

# Step 3: Solve for ( y ) and ( x )

Multiply (6) by -2 and add:

$$-2x - 8y + 2x + 3y = -8 + 10$$

$$-5y = 2 \Rightarrow y = -\frac{2}{5}$$

Substituting ( y ) into (6):

$$x = 4 + \frac{8}{5} = \frac{28}{5}$$

## Step 4: Solve for ( z )

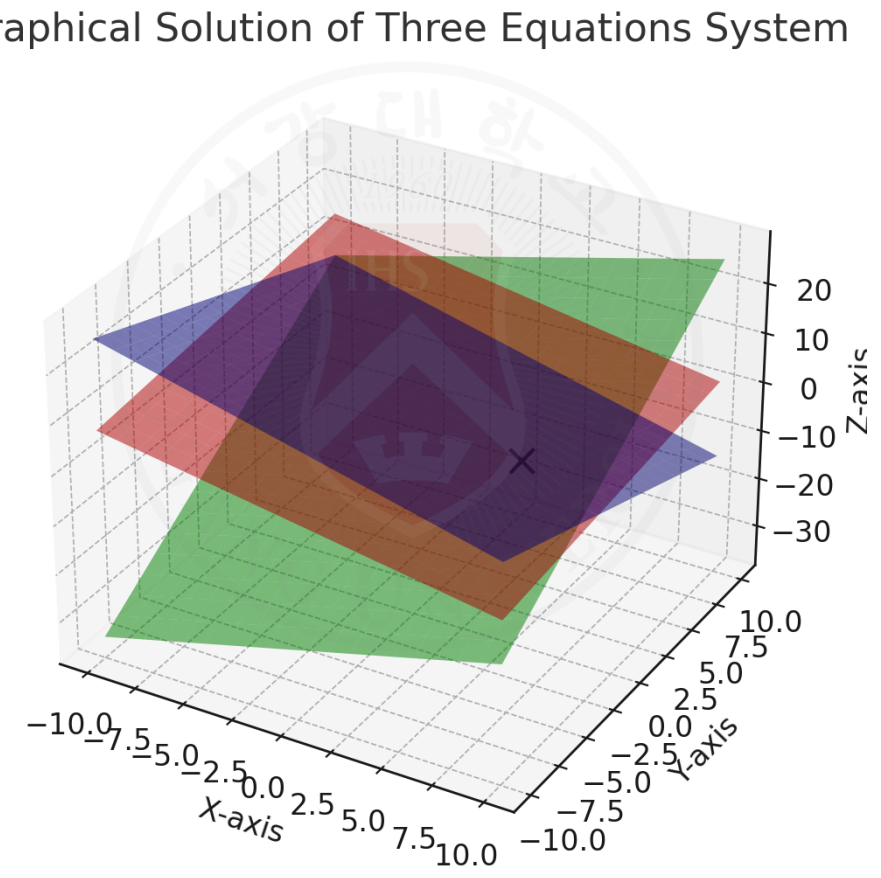
$$z = 3(-\frac{2}{5}) + 2 = \frac{4}{5}$$

## Final Answer

$$x = \frac{28}{5}, \quad y = -\frac{2}{5}, \quad z = \frac{4}{5}$$

# Graphical Solution of a 3x3 System of Equations

## Example:



# Your Turn: Practice Problems

## Please solve the following 3x3 systems of equations:

1. $( x + y + z = 6 )$, $( 2x - y + 3z = 14 )$, $( x + 2y - z = 4 )$
2. $( x + 2y + z = 6 )$, $( 2x - y + 3z = 14 )$, $( x + 2y - z = 4 )$

# Conclusion: Why This Math Matters

1. **Graphs of Equations**: Visual tools to understand relationships in economics.
2. **Solving Equations**: Essential for equilibrium analysis and optimization in economics.
3. **Systems of Equations**: Used to model complex economic relationships.
4. **Real-World Applications**: Informing policy decisions and business strategies.

# Next Steps

1. Practice **algebra problems from the textbook** (Jacques, Sections 1.3, 1.4).
2. Bring any questions to our **next class discussion!**

**Math is powerful—and fun!**

# Next Class

- (Mar 12) Supply and Demand Analysis (1.5), Transposition of Formulae (1.6), National Income Determination (1.7)

Any QUESTIONS?

Thank you for your attention!